

Condensate Binding, Charge Protection, and the Boundary of Particle Identification

Brad Wallace
Independent Researcher

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Abstract

We separate a tested classical binding mechanism from the identification of an elementary particle in the ACS condensate program. A specified coupled real–complex scalar model admits a qualified localized rotating solution with total energy below its free charged-wave threshold. Independent discretizations, tail behavior and energy identities support that conditional result. Explicit charge breaking produces leakage, while weak gauging retains some scalar–electric solutions but exposes omitted charged vector carriers. We then test proposed microscopic identifications involving framing, capacitance, Klein-cover maps, surface projection kernels, internal color states and Palatini torsion. Exact counterexamples invalidate several claimed implications while retaining the valid metric, representation and binding calculations. The result is a reproducible account of what the reviewed equations establish and which additional dynamical, normalization and quantum-state inputs are required for a physical mass or lifetime.

1 Introduction

A localized classical field, a conserved charge and an experimentally identified particle are distinct requirements. The aim here is to follow the ACS condensate proposal through those requirements without replacing a missing derivation by a fitted value or a change of terminology. The tested binding mechanism belongs to the Friedberg–Lee–Sirlin class, with a positive complex-field quartic added [1]. Its numerical qualification is useful even though its coefficients and physical units have not been selected by the microscopic proposal.

We first state the reduced model and its controls. We then include an existing gauge generator and its additional carriers. Finally, we test the geometric and action-level claims proposed to supply the missing physical identification. Exact algebraic obstructions are distinguished throughout from finite numerical observations and untested alternatives.

2 A specified self-binding model

Let χ be real and q complex. In the chosen dimensionless units,

$$E = \int_{\mathbb{R}^3} \left[\frac{|q_t|^2 + |\nabla q|^2 + \chi_t^2 + |\nabla \chi|^2}{2} + U(\chi, q) \right] d^3x, \quad (1)$$

$$U = \frac{(\chi^2 - 1)^2}{4} + \frac{\chi^2 |q|^2}{2} + \frac{b}{4} |q|^4, \quad Q = \int \text{Im}(\bar{q} q_t) d^3x. \quad (2)$$

The vacuum is $(\chi, q) = (1, 0)$ and the exterior charged-wave mass is one. For $q = f(r)e^{i\omega t}$ and $I = 4\pi \int r^2 f^2 dr$, the fixed-charge energy is

$$E_Q = G + V + \frac{Q^2}{2I}, \quad Q = \omega I. \quad (3)$$

Here G includes both field gradients. The condensate depletes where the charged field is large and creates its interior well dynamically; it is not a prescribed external barrier or a time-dependent energy source.

The inequality $E < |Q|$ prevents complete dispersal into the free charged waves of this model at fixed charge. It does not exclude all fission partitions, nonlinear disturbances or quantum decay channels. An exact control follows by restoring positive coefficients:

$$U - \frac{g^2 v^2 f^2}{2} = \frac{\lambda_\chi}{4} \left[(\chi^2 - v^2) + \frac{g^2}{\lambda_\chi} f^2 \right]^2 + \frac{1}{4} \left(b - \frac{g^4}{\lambda_\chi} \right) f^4. \quad (4)$$

Thus $b \geq g^4/\lambda_\chi$ implies $E \geq gv|Q|$; the tested $b = 1.2$ control cannot have the claimed binding advantage in our units.

2.1 Qualified numerical reference

For the chosen $b = 2\sqrt{3}/27$ and $Q = 1000$, an independent radial boundary-value solve gives

$$E = 838.163096, \quad \omega = 0.754888212, \quad \sqrt{1 - \omega^2} = 0.655853480. \quad (5)$$

The fitted exterior tail exponent is 0.655853649. The relative virial residual is 2.99×10^{-10} and the maximum boundary-value RMS residual is 9.91×10^{-8} . Finite-volume refinement from spacing 0.1 to 0.05 changes the energy from 838.131320 to 838.155154, consistent with the expected discretization trend.

The tested amplitude and phase sectors have no negative mode; the dipole mode converges toward the translation zero mode. Conservative dynamics and prepared-packet formation tests extend to dimensionless time 80. These are finite tests in the declared scalar model, not an asymptotic formation theorem or a full-theory particle lifetime. An initial optimizer failed its gradient criterion; the failed record and subsequent Newton qualification are both retained, without relaxing the criterion.

3 Charge breaking and gauge completion

In the full inherited scalar/Yukawa content, a generic uniform phase of Φ is not a symmetry. The allowed holomorphic Δ quartic breaks the candidate global X charge; even with that coefficient pair set to zero, weak mixed anomalies remain. The explicit reduced charge-breaking tests therefore measure leakage rather than assume an exact global protection. An independent radiation calculation gives the leading tested power $P \simeq 1.077996267 \epsilon^2$. A simulated finite charge-loss time is not an elementary-particle half-life.

For a gauged reduced model use $D_\mu q = (\partial_\mu - ieA_\mu)q$, $q = f(r)e^{i\omega t}$ and $\Omega = \omega - eA_0$. The equations are

$$\chi'' + 2\chi'/r = \chi(\chi^2 - 1 + f^2), \quad (6)$$

$$f'' + 2f'/r = (\chi^2 - \Omega^2)f + bf^3, \quad (7)$$

$$\Omega'' + 2\Omega'/r = e^2 f^2 \Omega. \quad (8)$$

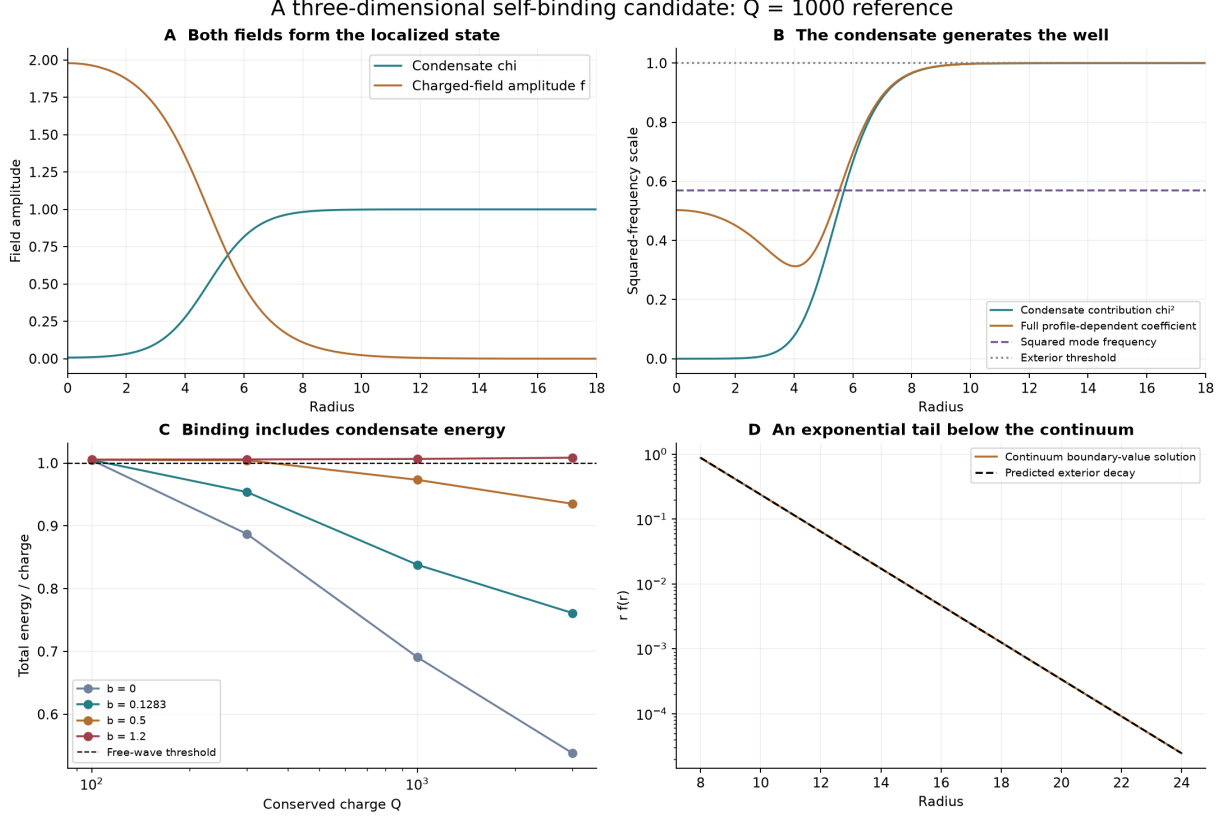


Figure 1: Conditional scalar binding, with all field energies included. The coefficient and charge are inputs; the tested profiles do not establish an elementary-particle identity.

With $A_0(\infty) = 0$, the exterior Coulomb contribution beyond radius R is

$$E_{\text{outside}} = \frac{e^2 Q^2}{8\pi R}, \quad \omega = \Omega(R) + R\Omega'(R). \quad (9)$$

Omitting this energy creates a false binding margin. At $Q = 1000, e = 0.12$, the total $E/Q = 0.96420948$ and $\omega = 0.96932000$; 14.323945 energy units lie outside the radius-40 domain. Independent finite-volume minimization, Gauss-law, virial and chemical-potential checks qualify the selected branch. Solver failures are retained and are not nonexistence proofs.

3.1 An existing gauge charge brings additional carriers

The scalar slice

$$\Phi = \chi I_2/2, \quad D_1 = D_2 = 0, \quad D_3 = qE_{44}/\sqrt{2}, \quad T^{15} = \frac{\text{diag}(1, 1, 1, -3)}{2\sqrt{6}} \quad (10)$$

has the reduced trace kinetic terms and $e = \sqrt{3/2}g_4$. A selected gauge-invariant potential reproduces the reduced scalar model. Its rank-one slice also has zero value and first derivative of the allowed holomorphic invariant, so no illegal single-component q^4 interaction has been silently removed.

However, the exterior $\Phi = I_2/2, \Delta = 0$ leaves $SU(4)$ unbroken. At unit gauge couplings the full 21-generator mass Gram has eighteen zeros and three eigenvalues $1/2$. Three complex color-lepton

vector pairs are massless and carry T^{15} weight of magnitude $2/3$ of the selected Δ component's weight. These classical modes invalidate using the massive scalar threshold alone as a full-theory stability certificate. No non-Abelian decay width or quantum confinement conclusion is inferred from this calculation.

The conventional neutral vacuum used in the companion action paper is a different background. Its proposed neutral component is electromagnetically neutral. Its charged partners have calculable spectra, but their masses and a condensate-to-particle map are not selected. Allowed Yukawa choices give opposite answers to fermion-emission threshold questions while obeying the same proposed dimensionless ratio. Additional carrier information is therefore a physical input, not a plotting choice.

4 Testing the proposed microscopic identifications

4.1 Framing does not determine the magnetic moment

The torus curve in the earlier note is a $(2, 1)$ embedding and an unknot. The spin-cover class contains a parity bit, whereas a gyromagnetic ratio is a real-valued dynamical observable. The repository had already withdrawn the identification of self-linking number with g ; this amendment applies that correction consistently to the papers that reused it.

For classical charge and mass moving along the same closed curve with uniform charge-to-mass ratio, magnetic moment and angular momentum are proportional to the same oriented vector area. Their conventional ratio gives $g = 1$, independently of winding or framing. Altering charge/current and mass distributions can change the moment while retaining the curve. This does not calculate the QED vertex or deny the electron's measured magnetic moment; it disproves the stated topological inference.

4.2 Capacitance retains a cutoff and a factor of two

Let $L = \log(8R/a) + 1$ and use the proposed

$$C = \frac{2\pi\epsilon_0 R}{L}, \quad q = \frac{e}{2}, \quad R = \frac{\hbar}{2mc}. \quad (11)$$

Then $E = q^2/(2C)$ gives

$$\frac{E}{mc^2} = \frac{\alpha L}{2}, \quad E = mc^2 \implies \alpha^{-1} = L/2. \quad (12)$$

The later archived CPF manuscript instead used $\alpha^{-1} = L$ and inserted $a/R = 8 \exp[-(K-1)]$. This produces any desired $L = K$ by construction. At its chosen $K = 137.035999171$, that exponential is 6.65896×10^{-59} ; its own self-energy equation requires $a/R = 8 \exp[-(2K-1)] \simeq 2.03905 \times 10^{-118}$ to reproduce the same inverse coupling. Neither cutoff follows from topology. The companion capacitance computations remain conditional model comparisons.

4.3 A cover map, a metric and a surface state are separate objects

For the source's orientation double cover, the deck action on first homology is $D = \text{diag}(1, -1)$. A lifted quotient diffeomorphism must commute with its unique nontrivial deck transformation. The proposed matrix fails this necessary condition:

$$M = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad MD - DM = \begin{pmatrix} 0 & -4 \\ 0 & 0 \end{pmatrix}. \quad (13)$$

This does not imply that identity cover homology determines a trivial mapping class; the earlier topology correction remains in force. The printed metric, by contrast, respects the glide pullback. Its determinant $R^2 + p^2 \cos^2(u/2)/(4\pi^2)$ retains independent radius and pitch inputs. A valid metric identity does not repair an invalid map.

Integrating a three-dimensional delta kernel over a two-dimensional surface produces a surface distribution. It does not by itself define a normalized volume L^2 state. A unit-integral transverse Gaussian of width ε has squared L^2 norm $1/(2\sqrt{\pi}\varepsilon)$, diverging as the width tends to zero. A smearing prescription, state domain and dynamics are required before the proposed projection can identify a particle.

4.4 Gauge-invariant statements cannot depend on real versus imaginary labels

The transformation $\psi \mapsto i\psi$ exchanges real and imaginary components while preserving $|D\psi|^2$. An imaginary component of a charged field does not thereby become electromagnetically dark. Likewise, the supplied Bell state and observables reproduce $2\sqrt{2}$, whereas the same settings on a product state give $\sqrt{2}$. The state and observables are assumptions, not consequences of a coordinate map. The manuscript's two-to-three coordinate map has rank two.

The color/music source proposes $(|R\rangle + |G\rangle + |B\rangle)/\sqrt{3}$ as a singlet. It is a vector in the fundamental triplet, not an invariant vector. Its two Cartan expectations vanish, but the sum of their variances is $1/3$ and its quadratic Casimir is $4/3$. Applying all eight generators gives a trivial common kernel in the fundamental representation. As positive controls, the triplet–antitriplet singlet and the antisymmetric three-triplet singlet are annihilated by the total generators. Zero mean charge and invariance are different conditions.

5 What the stated gravitational action supplies

For a smooth nondegenerate tetrad and independent metric-compatible connection, write $\omega = \omega_{LC} + K$. Then

$$R(\omega) = R(\omega_{LC}) + D_{LC}K + K \wedge K, \quad T = K \wedge e. \quad (14)$$

In the Palatini action, $D_{LC}(e \wedge e) = 0$ makes the term linear in $D_{LC}K$ a boundary term. Both $K \wedge K$ and an added $T^a \wedge *T_a$ term are algebraic in K . Thus the latter does not supply an independent contortion wave operator in this action. The component check retains all 24 independent contortion components. The same algebraic structure is exhibited in the quadratic-torsion decomposition of Ref. [2].

Curvature-derivative terms, singular defects or constrained teleparallel models require separate actions and mode analyses. They are not ruled out here, but their dynamics do not follow from renaming the algebraic term “kinetic.” The recovered soldering example likewise inserts a Lorentzian internal metric, a nondegenerate vierbein and a length scale. Its valid metric congruence preserves the supplied signature; it does not select it.

6 The RH claim has a separate, unresolved premise

The archived CPF curvature summand

$$\frac{2\varepsilon}{(1/2 - \beta)^2 - \varepsilon^2} \quad (15)$$

equals $-2/\varepsilon$ at $\beta = 1/2$, not zero. Independent finite-summand integration gives a vanishing long-time average for both on-line and off-line test locations away from contour poles. This neither justifies exchanging an infinite zero sum with a limit nor derives a map to physical cosmological curvature. No actual zeta zero is located or excluded by this countercheck. Finite ordinate statistics and physical stability analogies do not establish RH.

7 Discussion and conclusions

The surviving result is a conditional mechanism: a specified charged field can displace a condensate and bind below its scalar-wave threshold. The complete action determines whether additional fields change that conclusion. The recovered geometric proposals do not supply the missing normalization, charge sector, vacuum or quantum state. Several fail their own finite algebraic tests; others retain freely chosen dimensional data.

Further physical prediction needs a normalized microscopic action and condensate-to-field map, a justified coefficient and scale selection law, a physical vacuum, and a quantum identification of the carrier and its available decay channels. These requirements describe the boundary of the present derivation, not a prohibition on future ACS constructions.

A Reproducibility and retained qualifications

The scalar, charge-breaking and gauged data are preserved under `docs/condensate_energy/`. The endpoint package records the microscopic checks, the complete real roots of the historical quartic proxy, and the exact color-singlet controls. The companion action paper states the 68-component normalization and scale-family proof. The publication verification command is `pythonscripts/verify_publication.py`; its log distinguishes fresh portable checks from retained historical results and archive-only source-recovery tasks. Historical failures remain failures.

Acknowledgments and preparation

This paper consolidates the author’s research using AI-assisted drafting and computational checking. It does not assert external peer review or independent experimental confirmation of the microscopic model.

References

- [1] J. Heeck and M. Sokhashvili, “Revisiting the Friedberg–Lee–Sirlin soliton model,” arXiv:2303.09566, version 2 (2023).
- [2] C. Vashistha, R. Gannouji and A. Ganguly, “Neutron stars in Poincaré gauge gravity with quadratic torsion, arXiv:2606.09786, equations (4)–(10) and Sec. 2.2 (2026).